

improvements in semiconductor technology gradually tipped the balance in favour of transistors. By the 1960s, transistor-based computers had clearly emerged as the dominant technology, laying the foundation for the integrated circuits and modern microprocessors that power today's digital world. History suggests that the eventual winner among competing quantum architectures may likewise be difficult to predict, and the technology that ultimately defines the quantum age may not necessarily be the one that appears most promising today.

Now, let us look at some of the major competing quantum computing architectures. The ordering below roughly follows what I personally consider to be their present level of practicality and maturity, beginning with the most widely used and commercially successful approaches and ending with those that remain highly speculative. It should be emphasised that this ranking reflects a personal opinion informed by the general consensus within the quantum computing community and is by no means universally accepted. The eventual winner may well turn out to be very different from what many researchers expect today.

Please keep in mind that, in each of the quantum computing architectures listed below, the fundamental hardware problem being addressed is the physical realisation of qubits—that is, how qubits are formed, manipulated, and protected from errors.

- *Superconducting qubits* operate at temperatures of around 15 millikelvin, colder than outer space. This is currently the most mature and commercially successful approach, pursued by IBM, Google, Rigetti, and IQM.
- *Trapped-ion qubits* use actual atoms, such as ytterbium or barium ions, suspended and manipulated using electromagnetic fields and laser pulses. Major players include IonQ, Quantinuum, and Alpine Quantum Technologies.

- *Photonic qubits* use photons as carriers of quantum information and are among the few architectures that can operate at room temperature. Companies developing this technology include PsiQuantum, Xanadu, and ORCA Computing
- *Neutral-atom qubits* employ individual atoms trapped by optical tweezers and can be arranged into arbitrary two-dimensional and even three-dimensional patterns. Important companies in this area include QuEra Computing, Atom Computing, and Pasqal.
- *Silicon spin qubits* encode information in the spins of electrons confined in silicon and can potentially leverage existing semiconductor manufacturing processes. Significant efforts are underway at Intel, Diraq, and several academic laboratories.
- *Topological qubits* are theoretically expected to be much more resistant to errors than other qubit technologies, but many of the required physical phenomena remain to be conclusively demonstrated. This approach is being pursued primarily by Microsoft and various research groups.

At present, no one can say with confidence which of these architectures will ultimately lead to large-scale, fault-tolerant quantum computers. History teaches us that technological winners are often difficult to predict.

Measurement in quantum computing

Now, let us do some quantum programming. First, let us modify the quantum program we wrote in the previous article and observe the changes when it is executed. The modified Python program, titled '*qiskit2.py*' (with line numbers included for clarity), creates a single qubit and applies a Hadamard gate to it. However, unlike the previous version, this program also measures the state of the qubit.

```
1. from qiskit import QuantumCircuit
2. qc = QuantumCircuit(1, 1)
3. qc.h(0)
4. qc.measure(0,0)
5. print(qc.draw())
```

Now let us try to understand how this program works. Compared to the program '*qiskit1.py*' discussed in the previous article, Line 2 has been modified and Line 4 has been newly added.

Line 2 creates a quantum circuit containing one qubit and one classical bit. The first argument, 1, specifies the number of qubits in the circuit, while the second argument, also 1, specifies the number of classical bits available for storing measurement results. As before, the qubit is initialised in the state $|0\rangle$ by default. Line 4 performs a measurement operation on qubit 0 and stores the result in classical bit 0. The first argument, 0, denotes the qubit being measured, and the second argument, 0, denotes the classical bit in which the measurement outcome is recorded. After the measurement, the value stored in the classical bit will be either 0 or 1, depending on the outcome of the quantum measurement.

Recall that we installed Qiskit in a virtual environment named *qiskit_env*, which can be activated using the command '*conda activate qiskit_env*'. When the program '*qiskit2.py*' is executed with the command '*python qiskit2.py*', the output shown in Figure 1 is obtained. The output shows a visual representation of the quantum circuit generated by the program '*qiskit2.py*'. Compare this output with the one produced by '*qiskit1.py*', the program discussed in the previous article.

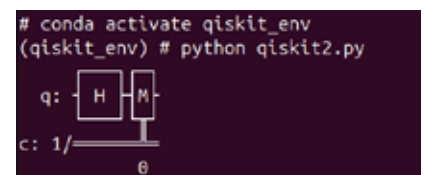


Figure 1: Output of the program *qiskit2.py*